Introduction to Wireless Communication Systems
LEC:1

Sub: **mobile communication systems**
Fourth class
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Reference
Wireless Communications: Principles and Practice,Theodore S. Rappaport

History of Wireless communication

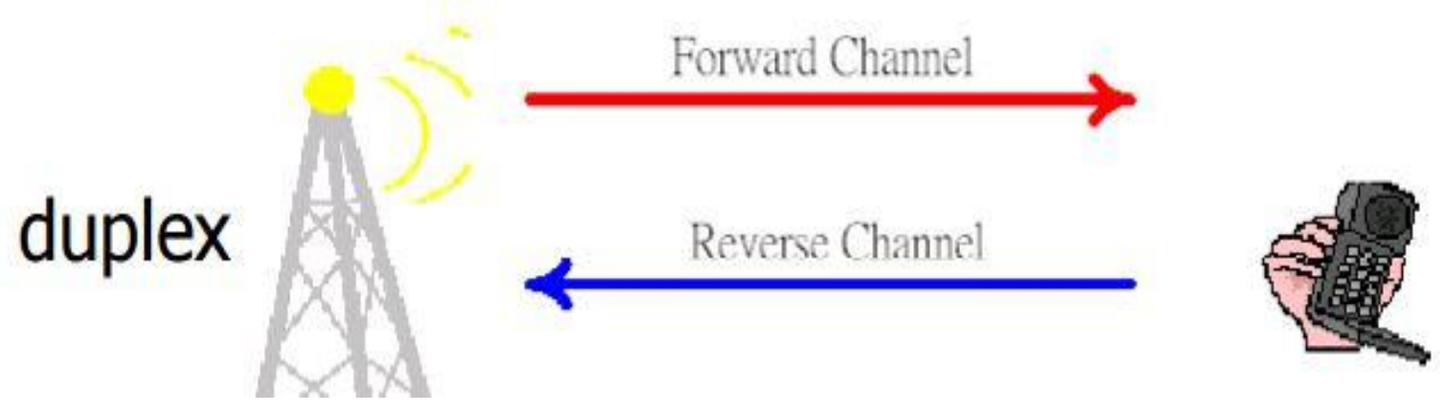
- 1899 - Marconi sends first radio message across Atlantic
- 1905 - Hulsmeyer detects ships with radar
- 1927 - US & Europe telephones linked by HF radio
- 1934 - AM mobile police radios for public safety widely used
- 1935 - Edwin Armstrong demonstrates FM radio system, became the primary modulation technique.
- 1940 - First microwave radar
- 1965 - First commercial communication satellite
- 1968 - AT&T proposes cellular phone system to Federal Communications Commission (FCC)
- 1983 - FCC allocates spectrum for analog cellular service (AMPS)
- 1990 - GSM digital cellular service introduced in Europe
- 1995 - FCC auctions new Personal Communication Service (PCS) licenses in U.S. for digital services
- 1998 - 40 million cellular phone users in U.S.
- 2000 - In some countries, mobile users outnumber conventional wireline customers.

Classification of mobile radio transmission system

- Mobile radio transmission systems may be classified as simplex, half-duplex or full-duplex.
- **simplex systems:** communication is possible in only one direction. Example **Paging** systems
- **Half-duplex**: radio systems allow two-way communication, but use the same radio channel for both transmission and reception. Example "**push-to-talk" and "release-to-listen**"
- **Full duplex systems**: allow simultaneous radio transmission and reception between subscriber and a base station, by providing two simultaneous but separate channels (frequency division duplex, or FDD) or adjacent time slots on a single radio channel (time division duplex, orTDD) for communication to and from the user.

Frequency division duplexing (FDD) & Time division duplexing (TDD)

- **Frequency division duplexing (FDD) :**uses two radio channel
- **forward channel** is The channel used to convey traffic from a base station to the MS
- **reverse channel :**The channel used to carry traffic from the MS to a base station
- FDD is used exclusively in analog mobile radio system



- **Time division duplexing (TDD):** uses the fact that it is possible to share a single radio channel in time, so that a portion of the time is used to transmit from the base station to the mobile, and the remaining time is used to transmit from the mobile to the base station

Types of wireless communication systems

- The major types of wireless communication systems are:
- 1- Paging Systems
- 2- Cordless Telephone Systems
- 3- Satellite communication systems
- 4- Wireless LAN systems
- 5- Cellular Telephone Systems

- **1.Paging systems are :** simplex communication systems that send brief messages to a subscriber. Depending on
- the type of service, the message may be either a numeric message, an alphanumeric message, or a voice message.
- Simultaneously broadcast paging message from each base station **(simulcasting)**
- **Large transmission power** to cover wide area, may cover a limited range **of 2 to 5 km**, or may even be confined to within individual buildings.
- **Reliable communication everywhere** (need good Signal to Noise performance).

Paging Systems

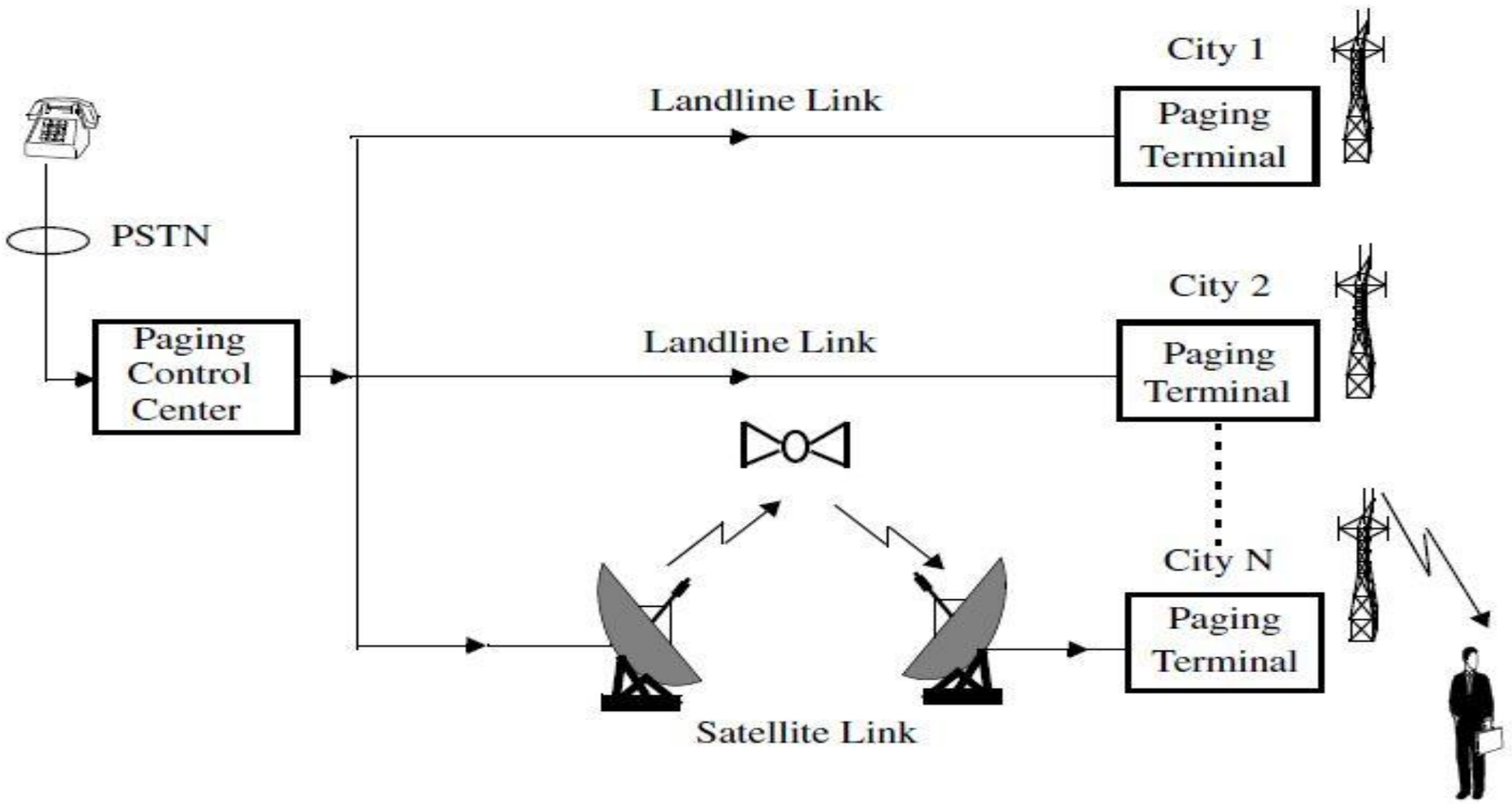


Figure 1.3 A wide area paging system. The paging control center dispatches pages received from the PSTN throughout several cities at the same time.

2. Cordless Telephone System

- Cordless telephone systems are full duplex communication systems that use radio to connect a portable handset to a dedicated base station, which is then connected to public switched telephone network (PSTN)..
- First generation cordless phone

1. in-home use (indoor)
2. few tens of meters

- Second generation cordless phone

1. outdoor
2. few hundred meters per station



Figure 1.4 A cordless telephone system.

3- Satellite communication Systems

The main feature of the satellite communication systems

1. Very wide range and coverage
2. Very useful in sparsely populated areas: rural areas, sea, mountains, etc.
3. Target: Vehicles and/or other stationary/mobile uses
4. Expensive base station (satellites) systems

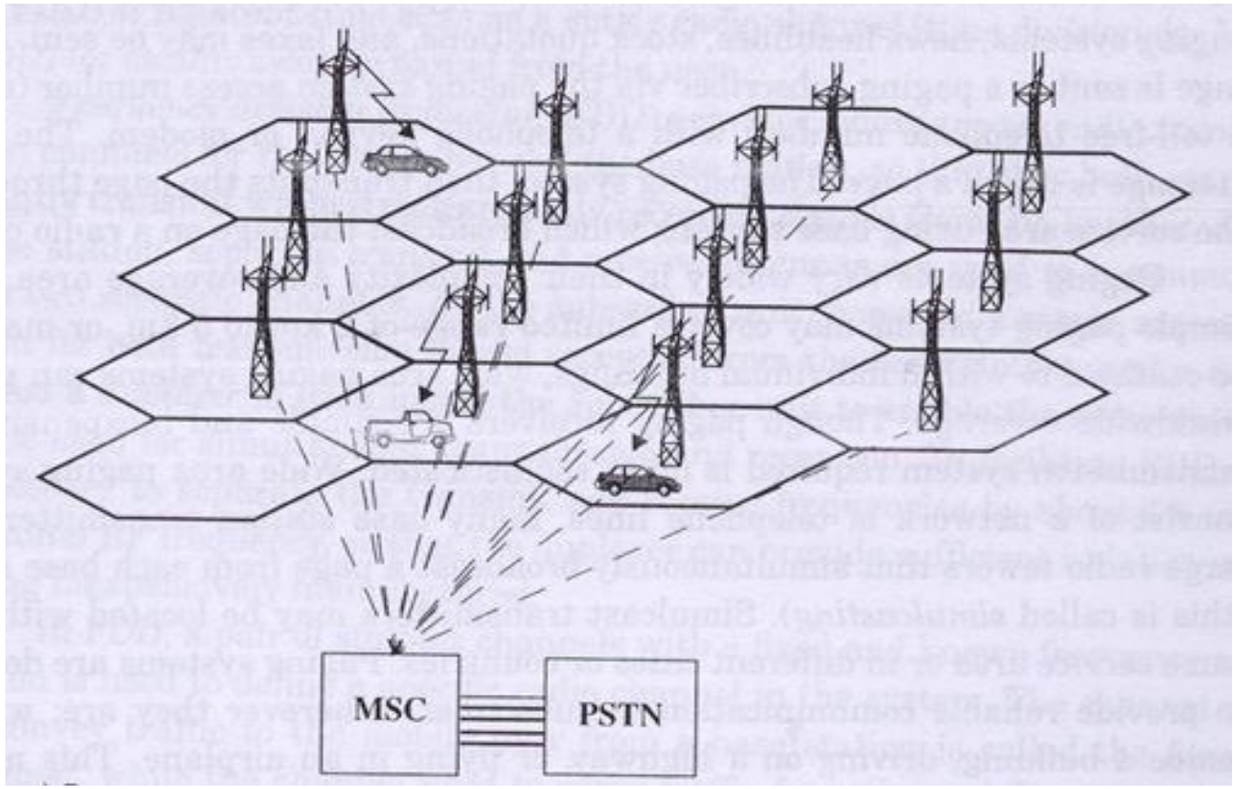
4- Wireless LAN (WLAN)

Characterized by

1. Low mobility (not for vehicular use)
2. High speed data transmission
3. Confined regions – buildings and campuses
4. Coverage: 100m – 300m per base station
5. Uses the following bands (902-928 MHz, 2400-2483.5 MHz, 5725-5850 MHz)

5.Cellular system

- Cellular radio is a technique that was developed to provide
  1. High capacity (large no. users)
  2. Large coverage area
  3. Efficient use of limited spectrum



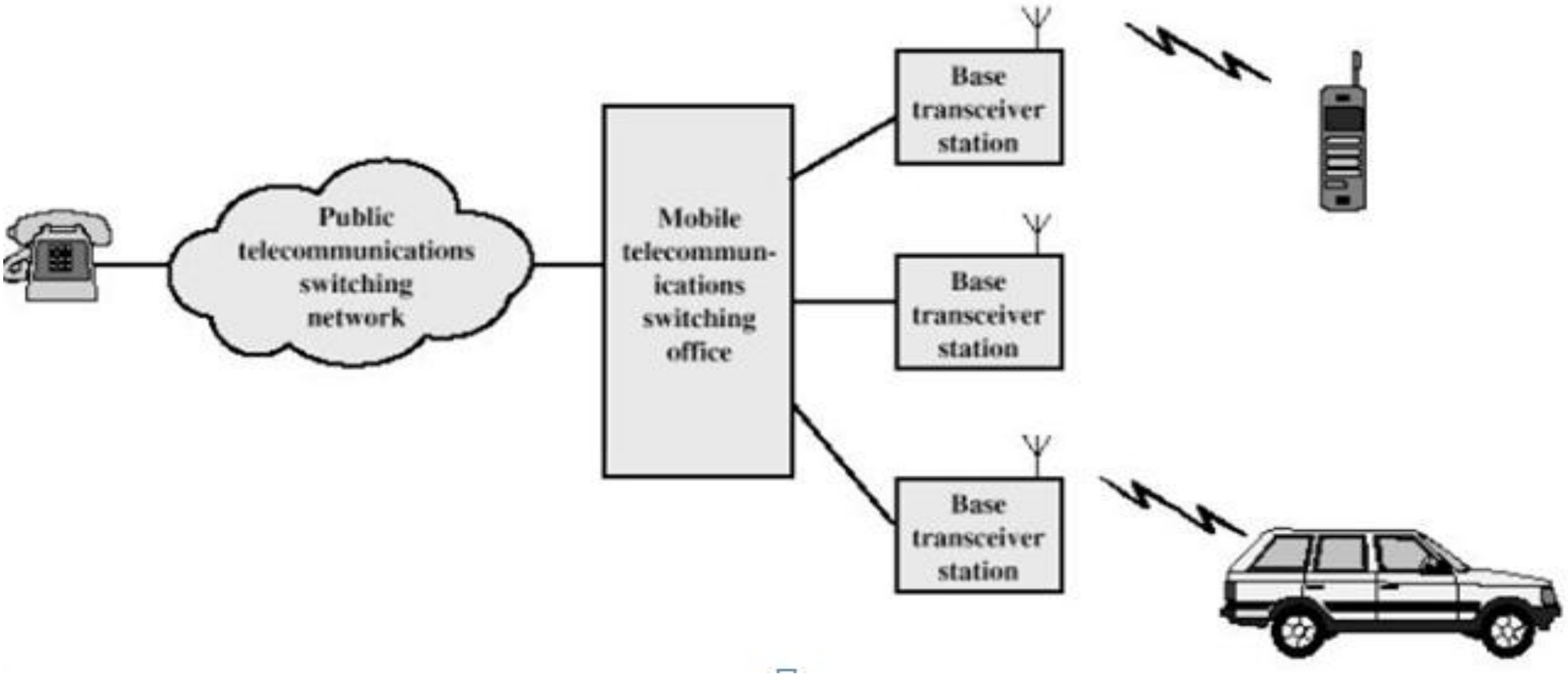
The main functions of the MTSO(MSC)

1. Assigns the voice channel to each call.
2. Performs handoffs.
3. Monitors the call for billing information.

<u>The main functions of the Base station</u>

1. serves mobile users in each cell
2. The base stations consist of several transmitters and receivers antennas which simultaneously handle full duplex communications and generally have towers .
3. serves as a **bridge** between all mobile users in the cell and connects the simultaneous mobile calls via telephone lines or microwave links to the MSC.

Cellular Systems Elements



Q/ What are the evolution parameters of cellular systems?

In order to offer higher **quality of service** Higher spectrum utilization efficiency.

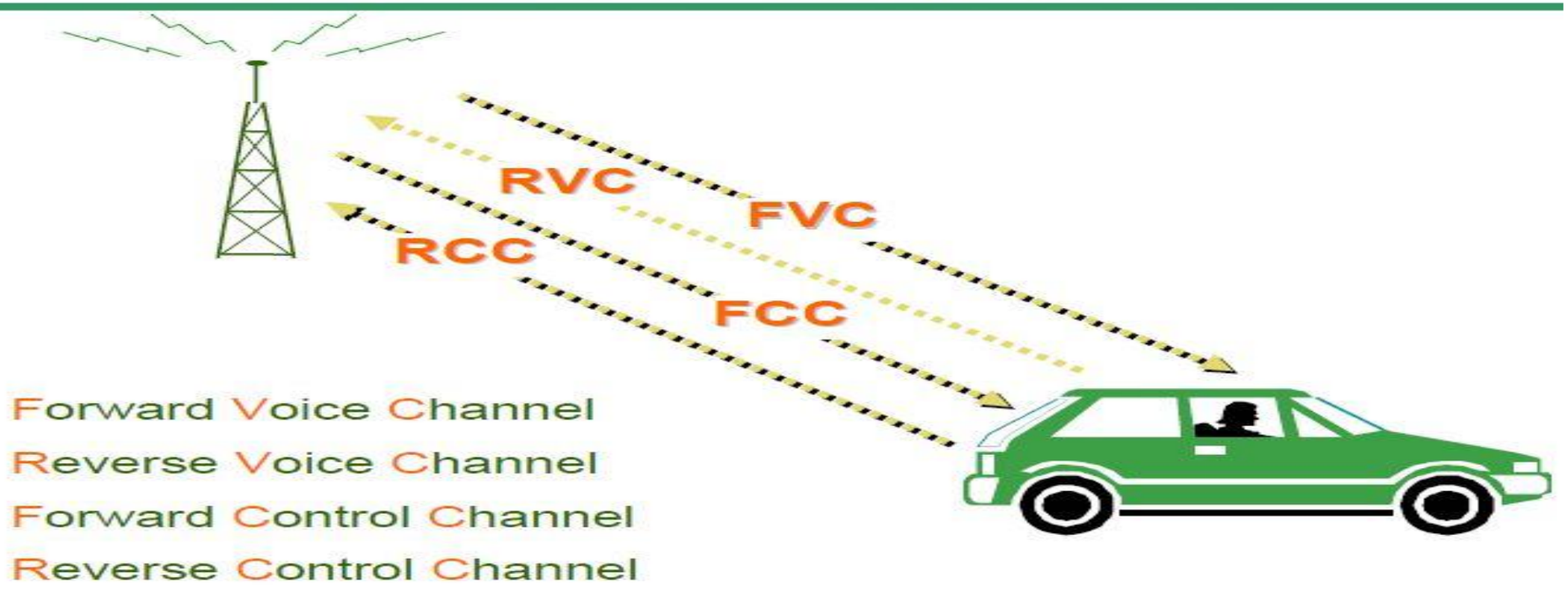
1. Improving coverage area
2. Increased capacity to support more subscribers (users).
3. Higher data rates.
4. Enhanced security
5. Lower service cost.
6. Offering new services such as Internet access and video calls.
7. Improved roaming capability.
8. **provide longer talking time.**

Q/ What types of transmissions in wireless communication systems ?

- 1.**Unicast** :(point- to –point) transmission is made from one device to a single other device. This means that the packet is addressed to one receiver
- 2. **Broadcast :** transmission from one device to all other devices
- One sender . the information is sent to all connected receivers
- 3.**Multicast** :transmission is made from one device to a subset of the other available devices

- Communication between the base station and mobiles is defined by the standard common air interface (CAI).
  - forward voice channel (FVC): voice transmission from base station to mobile
  - reverse voice channel (RVC): voice transmission from mobile to base station
  - forward control channels (FCC): initiating mobile call from base station to mobile
  - reverse control channel (RCC): initiating mobile call from mobile to base station

Base Station - Mobile Network



H.W

- Q/ What are the **Classifications of mobile radio transmission** system
- **Q/** Mention the **Types of wireless communication** systems
- Q/ What **types of transmissions** in wireless communication systems ?
- Q/ What are the operational channels used in each cell of the cellular system that take active parts during the call?